

MARIA GONZALEZ HERRERO

Software Engineer | Python, AI & Cybersecurity

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EXPERIENCE

Backend Developer

Humaind x Labs

12/2025 - Present León

- Development of backend services in Python and Django for a cybersecurity SaaS platform.
- Design and implementation of REST APIs for new features and internal integrations.
- Modelling of relational databases and query optimisation.
- Integration and management of the MISP platform for the automation of threat intelligence (CTI).

Programming, Robotics & 3D Printing Teacher

The Creative Minds Education

01/2024 - 06/2024 Madrid

- Delivered coding and robotics workshops using Python, Arduino, and 3D design tools.
- Introduced students to computational thinking and real-world tech applications.

3D Bio-printing Lab Assistant

ARIES, Universidad Nebrija

10/2023 - 03/2024 Madrid

- Designed and debugged scripts for automated GCODE generation of 3D biocomposite structures.
- Provided technical support for research in bio-printing workflows.
- Applied problem-solving skills to optimise code for experimental setups.

IT Risk Auditor

Santander Bank

07/2022 - 07/2023 Madrid

- Developed and implemented automation tools for complex IT risk audit processes.
- Supported the Technology Risk division in digitalising audits across multiple departments.
- Contributed to reducing manual audit workload and improving detection of IT vulnerabilities.

EDUCATION

MSc Computer Networks and Security

University of Essex

10/2024 - 10/2025 Colchester

Master's in Banking and Financial Markets

Universidad CUNEF

07/2022 - 06/2023 Madrid

BSc in Computer Engineering

Universidad Nebrija

09/2018 - 09/2023 Madrid

ABOUT ME

Computer Engineer with an MSc in Computer Networks & Security from the University of Essex. Back-end developer specialising in Python, Django and FastAPI, with experience in building SaaS platforms, designing REST APIs and developing applications used in production. Interested in scalable systems, automation and AI applied to software development.

SKILLS

<u>Python</u>	<u>Flutter</u>	<u>API</u>	<u>MongoDB</u>
<u>TypeScript</u>	<u>JavaScript</u>	<u>Postman</u>	
<u>MISP</u>	<u>PyTorch</u>	<u>LLMs</u>	
<u>Docker</u>	<u>Git</u>	<u>GraphQL</u>	
<u>MySQL</u>	<u>Model fine-tuning</u>	<u>Dart</u>	
<u>Firebase</u>	<u>Node.js</u>	<u>Image Segmentation</u>	
<u>Penetration tools</u>	<u>Django</u>		

PROJECTS

Glitch_ai - AI-Powered Pentesting Tool

Master's dissertation project developed in collaboration with **British Telecom (BT)** and the University of Essex, focusing on automated web security testing.

- Full-stack system (Python backend + Firebase authentication + Flutter frontend).
- Mistral AI integrated for vulnerability detection and the generation of intelligent mitigation reports.
- Integration of remote servers to circumvent system limitations for users without sudo privileges.

VOLUNTEERING

Emergency Support Volunteer

Hospital Fundación Jiménez Díaz

2019 - 2020 Madrid

Psychiatric Hospital Volunteer

Hermanas hospitalarias

2016 Palencia

AWARDS

 **TFG Excellence Award – Universidad Nebrija (2024)**
Recognised for outstanding thesis on algorithmic image segmentation for neurodegenerative disease detection.

PUBLICATIONS

González-Herrero, M., et al. “GLITCH_AI: A Hybrid Framework for Automated Penetration Testing with LLM-Driven Adaptation and Reporting”
[IEEE CAI 2026](#)

LINKS

[GitHub](#) [Portfolio](#) [LinkedIn](#)

PROJECTS

SanTelmo App - Field Notebook for SAT San Telmo

- Cross-platform Flutter app (iOS, Android, Web) for managing agricultural activities.
- Firebase-based cloud storage and user management system.
- Currently in use by SAT San Telmo for real-world field data tracking.

Neuroscience Project - U-Net Model

- Developed ML pipelines for dendritic spine detection in neuroimaging.
- Automated segmentation algorithms for neurodegenerative disease research.

CERTIFICATIONS

IELTS Certificate (English B2) – 2024

Diploma in Digital Transformation (IoT, Cloud, Big Data, Cibersecurity) – IBM (2019)

LANGUAGES

Spanish
Native 

English
B2 